

Customer Shopping Behaviour Analysis

A complete end-to-end Data Analyst project covering data cleaning, exploratory data analysis, SQL-based business problem solving, and Power BI dashboard creation – all aimed at identifying customer trends, purchasing patterns, and sales performance.

3,900

Total Customers

25

Unique Items

\$232,895.53

Total Sales

3.76

Avg. Review Rating

\$59.72

Avg. Purchase

About the Dataset

The dataset captures customer shopping behaviour across demographics, purchases, and preferences – enabling analysis of business performance and customer trends.



Demographics

Age, gender, location

Purchase Details

Category, item, amount, season

Behaviour

Frequency, payment method, subscription status

Feedback

Review ratings, color, size preferences

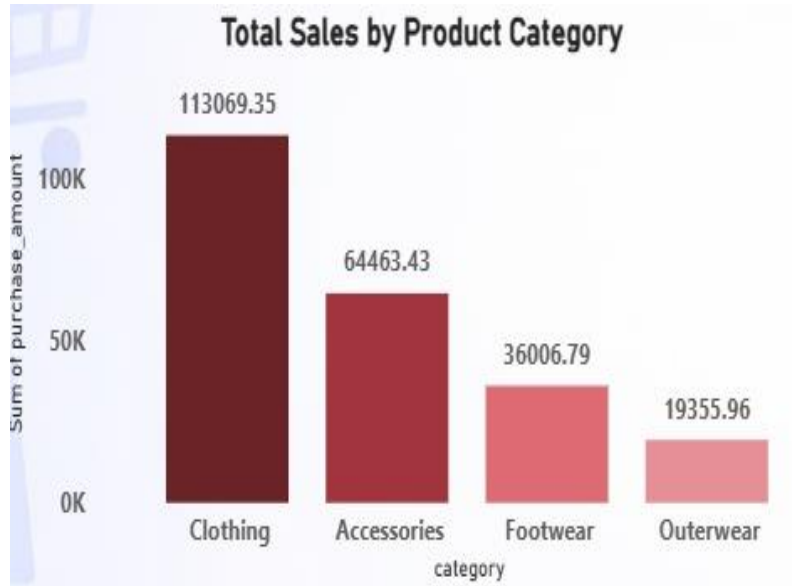
Data Cleaning with Python

Raw data contained missing values, duplicates, and formatting issues. Python & Pandas were used to prepare a clean, analysis-ready dataset.



Key techniques included category-based imputation for size, group-mean imputation for review ratings and purchase amounts, and boolean mapping for subscription and discount columns.

Sales Performance



Category Insights

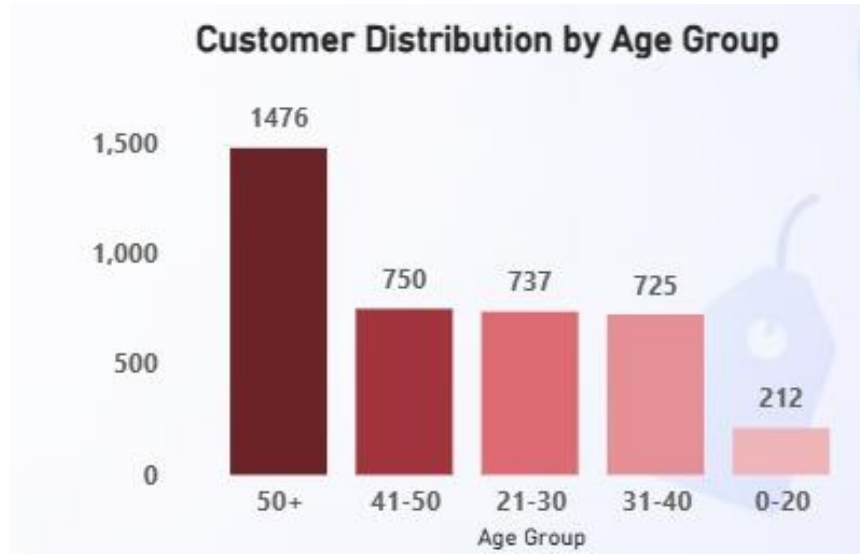
Clothing dominates with \$113K in sales — nearly half of total revenue. Outerwear lags significantly at \$19K.

- ✓ **Focus investment on Clothing; introduce discounts and fresh designs for Outerwear to close the gap.**

Seasonal Trend

Fall leads with \$59,521 in sales. Summer records the lowest at \$55,995 — a target for seasonal promotions.

Customer Demographics



Age Group Insights

Customers aged 50+ are the largest group (1,476) and the highest spenders (\$88,701). The 0-20 group is smallest with the lowest spend (\$12,524).

i Target older customers with premium products; use social media to attract the 0-20 segment.

Gender Split

Male customers account for 67.79% of total sales (\$157,875) vs. Female at 32.21% (\$75,020).

Geographic & Payment Insights

Top 5 Cities by Customer Count

Montana	96
California	95
Idaho	93
Illinois	92
Alabama	89

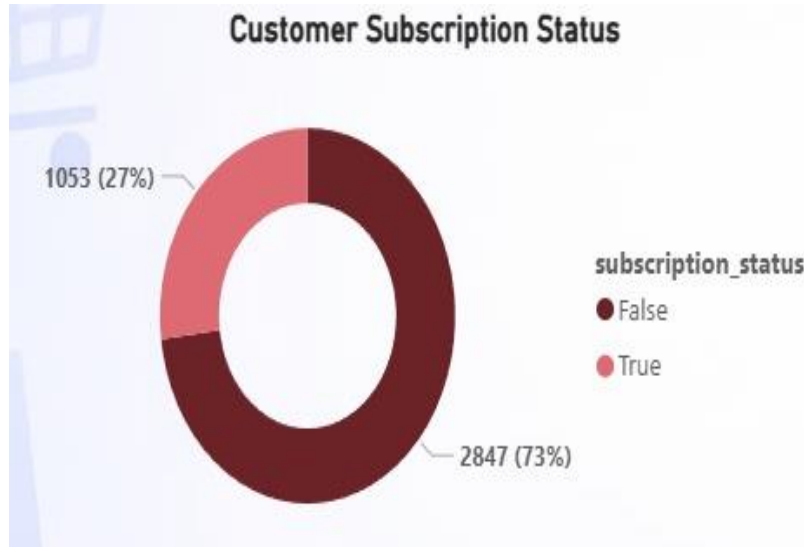
Preferred Payment Methods

PayPal	677 customers · \$40,223
Credit Card	671 customers · \$40,223
Cash	670 customers · \$39,827
Debit Card	636 customers · \$38,607
Venmo	634 customers · \$37,435
Bank Transfer	612 customers



PayPal & Credit Card lead in both usage and revenue —
prioritize digital payment rewards.

Subscriptions, Frequency & Preferences



Subscription Gap

Only 27% of customers hold active subscriptions. A major opportunity to grow loyalty and recurring revenue.

Purchase Frequency

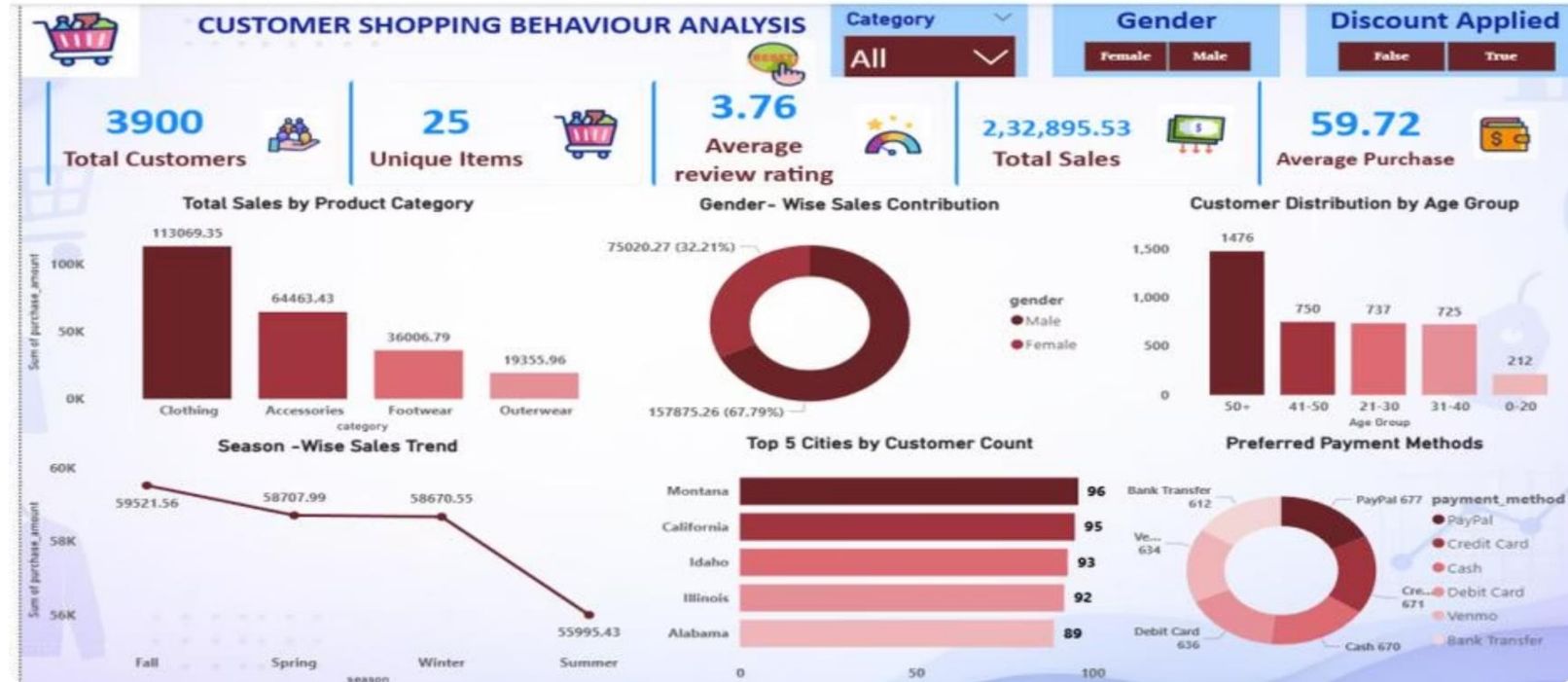
Every 3 Months is the most common pattern (584 customers), followed by Annually (572) and Quarterly (563) – indicating moderate repeat purchasing.

Size & Color Preferences

Medium (M) and Free Size dominate. Top colors: Olive (177), Yellow (174), Silver (173).

Power BI Dashboard

An interactive Power BI dashboard was built to visualize all key metrics and support business decision-making.





CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

Category

All

Gender

Female

Male

Discount Applied

False

True

3900

Total Customers



25

Unique Items



3.76

Average
review rating



2,32,895.53

Total Sales



59.72

Average Purchase



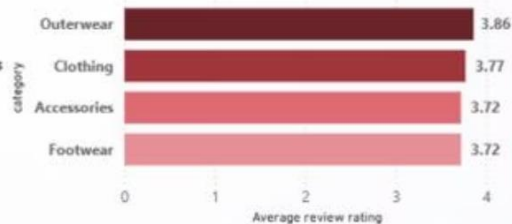
Customer Subscription Status



subscription_status

- False
- True

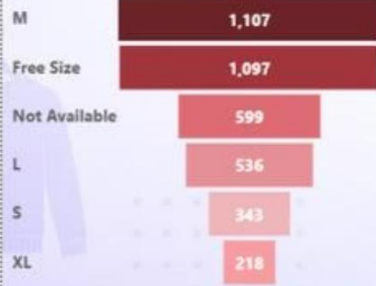
Average review rating by category



Customer Purchase Frequency Analysis

Every 3 Months	Monthly	Bi-Weekly
584		
Annually	553	547
572	Fortnightly	Weekly
Quarterly	542	539
563		

Most Preferred Product Sizes



Most Preferred Product Colors



Sales by payment method



Top Spending Age Groups



Key Business Insights

1

Clothing Leads

Highest-selling category; Outerwear has the lowest performance.

2

Male Dominance

Male customers drive ~68% of total sales revenue.

3

50+ Segment

Largest customer group and highest spenders (\$88,701).

4

Fall Peak

Fall records highest sales; Summer is the weakest season.

5

Low Subscriptions

Only 27% subscribed – a major loyalty growth opportunity.

6

PayPal Preferred

PayPal & Credit Card top both usage and revenue generation.



Recommendations & Conclusion

- **Boost Clothing & Fix Outerwear**
Increase Clothing stock; offer discounts and fresh designs for Outerwear.
- **Target High-Value Segments**
Premium strategies for 50+ customers; social media campaigns for the 0-20 group.
- **Grow Subscriptions & Loyalty**
Promote membership benefits, loyalty rewards, and repeat-purchase incentives.
- **Optimize Inventory**
Stock more Medium/Free Size products and Olive/Yellow color variants.

✓ This project successfully demonstrated the full Data Analyst workflow – from Python cleaning to SQL analysis to Power BI visualization – delivering actionable insights for customer targeting, inventory planning, and marketing strategy.

Tools Used: Python (Pandas, NumPy) · Jupyter Notebook · SQL · Power BI · CSV Dataset